

Homework 4

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Problem 1 Let G be a cumulative distribution function and suppose that for constants $a < b$, we wish to generate a random variable X with cumulative distribution function

$$F(x) = \frac{G(x) - G(a)}{G(b) - G(a)}, \quad a \leq x \leq b.$$

(a) If random variable Y has cumulative distribution function G , then under what information is F the conditional distribution of Y ? Prove your claim.

(b) Show that an acceptance-rejection method to generate X in this case is to generate a random variable Y having cumulative distribution function G and then accept it if it lies between a and b .

Solution:

(a) F would be the conditional distribution of Y if $a \leq Y \leq b$. This is because, for any $a \leq x \leq b$, we know that

$$\begin{aligned} \mathbb{P}(Y \leq x | a \leq Y \leq b) &= \frac{\mathbb{P}(Y \leq x; a \leq Y \leq b)}{\mathbb{P}(a \leq Y \leq b)} \\ &= \frac{\mathbb{P}(a \leq Y \leq x)}{\mathbb{P}(a \leq Y \leq b)} \\ &= \frac{G(x) - G(a)}{G(b) - G(a)}. \end{aligned}$$

The second equality is true since $x \leq b$.

(b) In the problem, we want to show that we can use an acceptance-rejection method to generate X using a proposal distribution G with some appropriate constant C where we accept a value if it falls between a and b ; otherwise, we will reject it.

First, we can derive the pdf of X as follows:

$$f(x) = \frac{d}{dx} F(x) = \frac{g(x)}{G(b) - G(a)} \mathbb{1}_{\{a \leq x \leq b\}}(x).$$

For the A-R method, we use the proposal density g of the distribution G . Then, we select the constant C to be $\frac{1}{G(b) - G(a)}$. We can verify this choice of C by showing that

$$\frac{f(x)}{g(x)} = \frac{\frac{g(x)}{G(b) - G(a)} \mathbb{1}_{\{a \leq x \leq b\}}(x)}{g(x)} = \frac{\mathbb{1}_{\{a \leq x \leq b\}}}{G(b) - G(a)} \leq \frac{1}{G(b) - G(a)} = C.$$

Thus, the A-R will accept a generated value x with probability

$$\frac{f(x)}{Cg(x)} = \frac{\frac{g(x)}{G(b) - G(a)} \mathbb{1}_{\{a \leq x \leq b\}}(x)}{\frac{1}{G(b) - G(a)} g(x)} = \mathbb{1}_{\{a \leq x \leq b\}}(x).$$

In other words, we accept x with probability one if x is contained in $[a, b]$; otherwise, we reject x .

Problem 2 Derive an acceptance-rejection method to generate (X, Y) with joint density function

$$f(x, y) = \frac{1}{K} \exp(-x^2 - y^2 + x \sin(xy)), \forall x, y \in \mathbb{R},$$

where K is a normalizing constant. Implement your method in a computer to generate 100 pairs of (X, Y) , and plot the distributions of X and Y respectively.

Solution:

We want to find a proposal distribution $g(x, y)$ such that

$$\frac{\exp(-x^2 - y^2 + x \sin(xy))}{g(x, y)} \leq C$$

for all $x, y \in \mathbb{R}$ and for some fixed positive constant C .

Recall lecture examples, one candidate for simulating (X, Y) we might consider is a joint exponential random variable with density $g(x, y) = e^{-x}e^{-y}$ for $x, y \geq 0$; however, its support does not contain negative values, unlike $f(x, y)$. Therefore, we can consider a distribution similar to the exponential distribution, namely the joint Laplace distribution with the following density

$$g(x, y) = \left(\frac{1}{2}e^{-|x|}\right) \left(\frac{1}{2}e^{-|y|}\right).$$

Note that X and Y here are independent, so we can simulate them separately.

Then, we will show, with this proposed distribution, the ratio is bounded.

$$\begin{aligned} \frac{\exp(-x^2 - y^2 + x \sin(xy))}{g(x, y)} &= \frac{4 \exp(-x^2 - y^2 + x \sin(xy))}{\exp(-|x| - |y|)} \\ &= 4 \exp(-x^2 - y^2 + x \sin(xy) + |x| + |y|) \\ &= 4 \exp(-(x^2 - 2|x| + 1) - (y^2 - 2|y| + 1) + (x \sin(xy) - |x|) - |y| + 2) \\ &= 4 \exp(-(|x| - 1)^2 - (|y| - 1)^2 + (x \sin(xy) - |x|) - |y| + 2) \\ &\leq 4e^2. \end{aligned}$$

The inequality is true because the following terms is non-negative (i.e., $(|x| - 1)^2 \geq 0, (|y| - 1)^2 \geq 0, |y| \geq 0$). While the third term, $(x \sin(xy) - |x|) \leq |x| - |x| \leq 0$, by sine function property.

Therefore, we can simulate (X, Y) with density f by (i) simulating (X', Y') with density $g(x, y)$ and (ii) accepting (X', Y') with probability

$$\frac{\exp(-x^2 - y^2 + x \sin(xy))}{4e^2 (\exp(-|x| - |y|)/4)} = \exp(-x^2 - y^2 + x \sin(xy) + |x| + |y| - 2).$$

Assume that we can only access a uniform random sampler. Then, we can implement the above as follows:

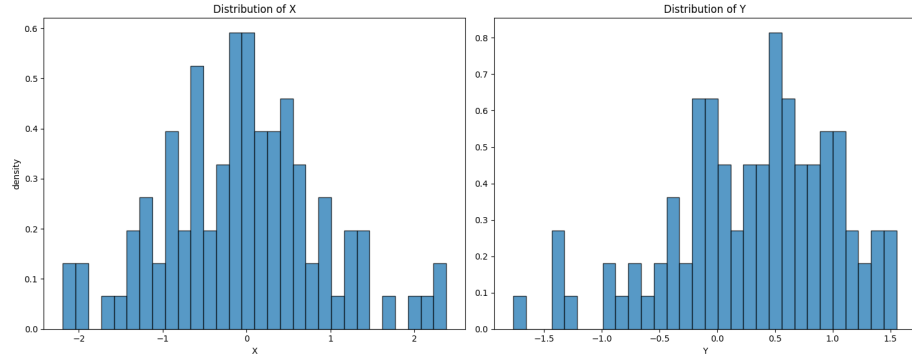
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1 while True do
2   Simulate  $U_x \leftarrow \text{rand}(0, 1)$ ;
3   Simulate  $V_x \leftarrow \text{rand}(0, 1)$ ;
4   Simulate  $U_y \leftarrow \text{rand}(0, 1)$ ;
5   Simulate  $V_y \leftarrow \text{rand}(0, 1)$ ;
6   Simulate  $U \leftarrow \text{rand}(0, 1)$ ;
7   if  $V_x \geq 0.5$  then
8     Set  $X \leftarrow -\log(1 - U_x)$ ;
9   else
10    Set  $X \leftarrow \log(1 - U_x)$ ;
11   if  $V_y \geq 0.5$  then
12     Set  $Y \leftarrow -\log(1 - U_y)$ ;
13   else
14     Set  $Y \leftarrow \log(1 - U_y)$ ;
15   if  $U \leq \exp(-X^2 - Y^2 + X \sin(XY) + |X| + |Y| - 2)$  then
16     return  $(X, Y)$ 

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One potential difficulty is to generate a Laplace-distributed r.v. X . Observe that it is symmetric, hence we first generate an exponential distribution using the ITM by setting $X' = -\log(1 - U_x)$. Then, we either pick $X = X'$ or $X = -X'$ with equal probabilities of 0.5. This is similar for generating Y . Then, we accept (X, Y) with probability $\exp(-X^2 - Y^2 + X \sin(XY) + |X| + |Y| - 2)$. We restart the entire process if the algorithm does not accept (X, Y) .

The sample distributions for X and Y are presented here:



Problem 3 Derive a procedure to sample uniformly from the region

$$A = \{(x, y) : x \in [-1, 1], |y| \leq |x|^{-1/2}\}.$$

Implement your procedure in a computer to generate 100 pairs of (X, Y) , and plot their positions on a two-dimensional plane to check that they indeed appear uniformly distributed over the region A .

Solution:

In class, we see that there are two ways to simulate uniformly over an irregular region: a naive approach and an alternative approach that simulates the marginal distribution of X directly.

In this case, the naive approach would fail since we could not find a value K used in the algorithm (as y is unbounded as $x \rightarrow 0$). Therefore, we will look at the alternative approach covered in class.

Before going into details, we can simplify by noticing that the region A is symmetric along both the horizontal and the vertical axis, so we can first concentrate on the area in the first quadrant, denoted as

$$A_1 = \{0 \leq x \leq 1, 0 \leq y \leq h(x)\}$$

where $h(x) = x^{-1/2}$.

To uniformly simulate over A_1 , we follow these three steps:

Step 1: Sampling X :

We note that X is sampled in proportion to its density. Writing its density as $f(x) = \frac{h(x)}{K}$, we can find K by solving

$$K = \int h(x)dx = \int_0^1 x^{-\frac{1}{2}}dx = 2.$$

Therefore, the density of X is $f(x) = \frac{x^{-\frac{1}{2}}}{2}$. CDF can be derived as follows:

$$F(x) = \mathbb{P}(X \leq x) = \int_0^x \frac{y^{-\frac{1}{2}}}{2}dy = x^{\frac{1}{2}}, x \in [0, 1].$$

Therefore, we can use the ITM to simulate X by sampling $U \sim Unif(0, 1)$ and then applying $X = F^{-1}(U) = U^2$. Alternatively, one can use the acceptance-rejection method without calculating the normalizing constant K . This requires finding an appropriate proposal density g , finding a constant $\tilde{C}$ such that $\frac{h(x)}{g(x)} \leq \tilde{C}$, and then accepting a sample with probability $\frac{h(x)}{\tilde{C}g(x)}$.

Step 2: Sampling Y

For any given X , Y has a uniform distribution over its range $[0, h(X)]$. Therefore, to generate Y , we simulate $V \sim Unif(0, 1)$ and set $Y = V \cdot h(X) = V[U^2]^{-\frac{1}{2}} = VU^{-1}$. Note that U and V are independent.

Step 3: Expanding from A_1 to A

For any $(X, Y) \in A_1$, we know that these four points— (X, Y) , $(-X, Y)$, $(-X, -Y)$, and $(X, -Y)$ —are equally likely. So, we can expand our results by simulating $W \sim Unif(0, 1)$ and then selecting

$$(X^*, Y^*) = \begin{cases} (X, Y) & W \in [0, \frac{1}{4}); \\ (-X, Y) & W \in [\frac{1}{4}, \frac{2}{4}); \\ (-X, -Y) & W \in [\frac{2}{4}, \frac{3}{4}); \\ (X, -Y) & W \in [\frac{3}{4}, 1]. \end{cases}$$

The entire algorithm for generating one sample is summarized as follows:

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1  $U \leftarrow \text{rand}(0, 1)$  ;
2  $V \leftarrow \text{rand}(0, 1)$  ;
3  $X \leftarrow U^2$  ;
4  $Y \leftarrow \frac{V}{U}$  ;
5 return  $(X, Y)$ 

```

To convert each point (X, Y) in A_1 to (X^*, Y^*) in A , we can do the following:

```

1  $W \leftarrow \text{Unif}(0, 1)$  ;
2 if  $W < 0.25$  then
3   return  $(X, Y)$ 
4 else if  $W < 0.5$  then
5   return  $(-X, Y)$ 
6 else if  $W < 0.75$  then
7   return  $(-X, -Y)$ 
8 else
9   return  $(X, -Y)$ 

```

The plot is shown below.

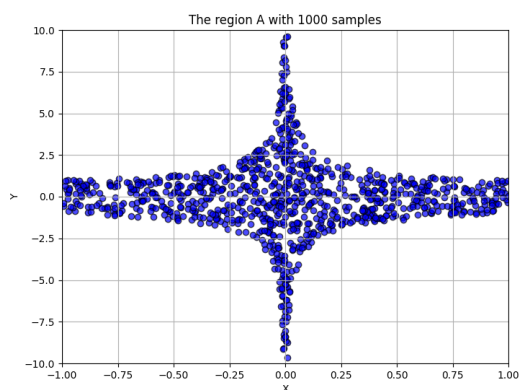


Figure 1: We simulated 1000 samples to make the visual clearer.

Problem 4 Consider the discrete random variable X where

$$\mathbb{P}(X = j) = \left(\frac{1}{2}\right)^{j+1} + \left(\frac{1}{2}\right) \frac{2^{j-1}}{3^j}, j = 1, 2, \dots$$

which appeared in HW2 Problem 3. Instead of using the inverse transform method as there, now present a composition method to generate X . Then implement your procedure in a computer to generate 100 copies of X , and plot their distribution.

Solution:

We notice that

$$\mathbb{P}(X = j) = \left(\frac{1}{2}\right)^{j+1} + \left(\frac{1}{2}\right) \frac{2^{j-1}}{3^j} = \frac{1}{2} \left(\frac{1}{2}\right)^{j-1} \left(\frac{1}{2}\right) + \frac{1}{2} \left(\frac{2}{3}\right)^{j-1} \left(\frac{1}{3}\right).$$

This is a mixture distribution of two pmfs of geometric random variables with rates $\frac{1}{2}$ and $\frac{1}{3}$.

Concretely, we want to simulate X as a J -mixture random variable of Y_1 and Y_2 where $\mathbb{P}(J = 1) = 0.5$, $\mathbb{P}(J = 2) = 0.5$, $Y_1 \sim \text{Geom}(1/2)$, and $Y_2 \sim \text{Geom}(1/3)$. We know that

$$\mathbb{P}(Y_1 = k) = \left(\frac{1}{2}\right)^{k-1} \left(\frac{1}{2}\right)$$

and

$$\mathbb{P}(Y_2 = k) = \left(\frac{2}{3}\right)^{k-1} \left(\frac{1}{3}\right).$$

To verify this equivalency, we have that, by the law of total probability,

$$\begin{aligned} \mathbb{P}(X = k) &= \mathbb{P}(J = 1)\mathbb{P}(Y_1 = k) + \mathbb{P}(J = 2)\mathbb{P}(Y_2 = k) \\ &= \frac{1}{2} \left(\frac{1}{2}\right)^{k-1} \left(\frac{1}{2}\right) + \frac{1}{2} \left(\frac{2}{3}\right)^{k-1} \left(\frac{1}{3}\right) \\ &= \left(\frac{1}{2}\right)^{k+1} + \left(\frac{1}{2}\right) \frac{2^{k-1}}{3^k}. \end{aligned}$$

J is either 1 or 2 with equal probabilities. If $J = 1$, then we will simulate $Y_1 \sim \text{Geom}(1/2)$ and return $X = Y_1$; if $J = 2$, then we simulate $Y_2 \sim \text{Geom}(1/3)$ and return $X = Y_2$.

Assume that we only have access to a random generator for $\text{Unif}(0, 1)$. We can implement the composition method as follows:

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1  $U \leftarrow \text{rand}(0, 1);$ 
2  $V \leftarrow \text{rand}(0, 1);$ 
3 if  $U \leq 0.5$  then
4   return  $\left\lceil \frac{\log(1-V)}{\log(1-1/2)} \right\rceil;$ 
5 else
6   return  $\left\lceil \frac{\log(1-V)}{\log(1-1/3)} \right\rceil;$ 
```

That is, $J = 1$ when $U \leq 0.5$ with probability 0.5, and we return Y_1 using the inverse transform method (for $\text{Geom}(1/2)$) in the following formula: $\left\lceil \frac{\log(1-V)}{\log(1/2)} \right\rceil$. Similarly, $J = 2$ when $U > 0.5$ with probability 0.5, and we return Y_2 using the inverse transform method (for $\text{Geom}(1/3)$) in the following formula: $\left\lceil \frac{\log(1-V)}{\log(2/3)} \right\rceil$.

The simulated distribution is shown as follows:

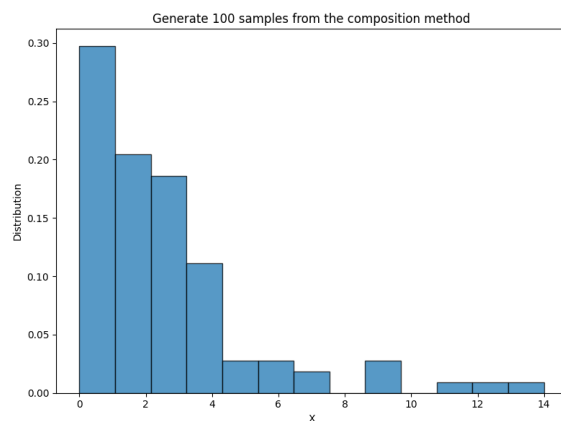


Figure 2: This is generated from the composition method

We can compare this to our plot from HW2 presented below:

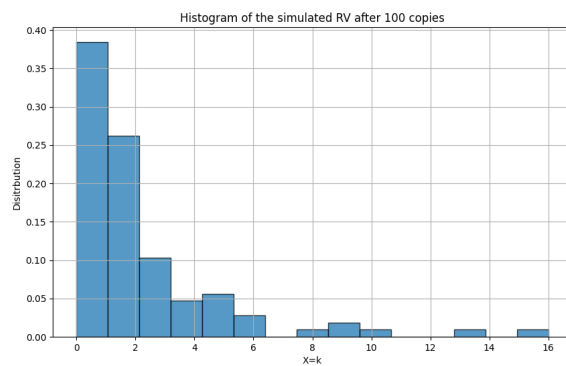


Figure 3: This is generated from the ITM

Problem 5 Present a method to generate a random variable X having cumulative distribution function

$$F(x) = \int_0^\infty x^y e^{-y} dy, 0 \leq x \leq 1.$$

Implement your procedure in a computer to generate 100 copies of X , and plot their distribution.

Solution:

To generate this random variable X with this given cdf, we will use the composition method for a mixture of a continuum set of random variables. In short, we show that we can independently sample x, y , and treat y as the mixture weights, and achieve the same underlying cdf.

Specifically, we aim to generate X from a random variable Y_T with a random index T , in which the cdf of Y_t is $\mathbb{P}(Y_t \leq x) = x^t$ for $0 \leq x \leq 1$, and the density of T is $f_T(t) = e^{-t}$ for $t \geq 0$. This Y_T is equal to our targeted X in distribution since their cdfs are the same, i.e.,

$$\mathbb{P}(Y_T \leq x) = \int_0^\infty \mathbb{P}(Y_t \leq x) f_T(t) dt = \int_0^\infty x^t e^{-t} dt = F(x).$$

We note that $\mathbb{P}(Y_t \leq x) = x^t$ is, in fact, a valid cdf since it is non-decreasing over $[0, 1]$, and $\mathbb{P}(Y_t \leq 1) = 1^t = 1$, and $\mathbb{P}(Y_t \leq 0) = 0^t = 0$.

Also, $f_T(t) = e^{-t}$ is the density of an exponential random variable with parameter 1.

To generate a sample of this random variable X , we first generate T via the inverse transform method. Specifically, we take $U \sim \text{Unif}(0, 1)$ and then apply $T = -\ln(1 - U)$. For a sample T , we want to sample random variable Y_T with a cdf $F_{Y_T}(x) = x^T$. This again can be simulated via ITM. That is, given T , we take $V \sim \text{Unif}(0, 1)$ and then apply $X = V^{1/T}$.

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- 1 Simulate $U \leftarrow \text{rand}(0, 1)$;
 - 2 Simulate $V \leftarrow \text{rand}(0, 1)$;
 - 3 Set $T \leftarrow -\ln(1 - U)$;
 - 4 Set $X \leftarrow V^{1/T}$;
 - 5 **return** X
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The plot is illustrated below.

